AyuAware Chat History Section

Technical Product Requirements Document

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1. Executive Summary

What We're Building

The Chat History section is the **Conversation Archive & Discovery Hub** — where users find, manage, and revisit their conversations with the Digital Twin. It serves three primary purposes:

1. **Finding specific conversations** — "Where did we discuss my sleep issues?"
2. **Resuming ongoing threads** — Continue where you left off
3. **Reviewing journey/progress** — See evolution over time

This isn't a basic chat log. It's a sophisticated discovery and management interface that:

Dimension	What We Offer	Why It Matters
Smart Categorization	Conversations grouped by type (Check-ins, Deep Dives, Chats)	Easy to find what you're looking for
Flexible Discovery	Multiple ways to find conversations (chronological, topics, search)	Different needs, different approaches
Semantic Search	Natural language queries ("When did we discuss my stress?")	Find conversations the way you think
Action-Rich	Full conversation management (pin, favorite,	Complete control over your
Topic	Conversations grouped by health themes	See patterns and progress by
Practitioner	Share conversations with healthcare providers	Bridge digital and

Why It Matters

Users build rich conversation histories with their Digital Twin — morning check-ins, deep constitutional analyses, casual questions, protocol guidance. Without good organization and discovery, this valuable content becomes inaccessible.

The Chat History section ensures users can:

- Find that specific conversation where they discussed something important
- Resume multi-session discussions without losing context
- Review their journey and see how their health has evolved
- Share relevant conversations with practitioners
- Export conversations for personal records

Key Differentiator

Generic chat app history: "List of messages. Scroll to find. Delete."

AyuAware Chat History: "Your conversations organized by type (Check-ins, Analyses, Chats) and topic (Sleep, Digestion, Stress). Search naturally: 'When did we discuss my sleep issues?' Pin important conversations. Share with your practitioner. Export for your records. Continue any thread seamlessly."

Intelligent organization. Natural search. Full management. Journey visibility.

2. Section Overview

2.1 Information Architecture

```
Chat History Section
│
├── 📋 Header
│    ├── Back Navigation
│    ├── Title
│    ├── View Mode Toggle (List/Grid)
│    └── Settings
│
├── 🔍 Smart Search
│    ├── Natural Language Search
│    ├── Keyword Search
│    └── Filter Pills (when active)
│
├── 📅 Chronological View (Default)
│    ├── Today
│    ├── Yesterday
│    ├── Last 7 Days
│    ├── Last 30 Days
```

```
│       └── Older
│
├── 🏷️ Topic-Based View (Toggle)
│       ├── Sleep & Recovery
│       ├── Digestion & Nutrition
│       ├── Stress & Mental Health
│       ├── Energy & Vitality
│       ├── Constitution & Balance
│       ├── Symptoms & Concerns
│       ├── Protocols & Practices
│       └── General
│
├── 💬 Conversation Cards
│       ├── Category Icon & Color
│       ├── Title
│       ├── Timestamp
│       ├── Type Badge
│       ├── Status Indicators
│       └── Actions (swipe/long-press)
│
├── 📌 Pinned Conversations
│       └── Always at Top
│
└── ⭐ Favorited Conversations
        └── Marked for Quick Access
```

2.2 Conversation Taxonomy

Three Main Categories:

Categor	Icon	Color	Badge	Examples
Check-ins	✓ (checkmark)	#4A9B8E (teal)	"Check-in"	Morning/evening routines, progress tracking, structured assessments
Deep Dives	🔍 (magnifying	#E89B7F (coral)	"Analysis"	Constitutional analysis, health metrics analysis, protocol generation
Casual Chat	💬 (chat bubble)	#7AB8AC (light teal)	"Chat"	General questions, quick advice, informal conversations

2.3 View Modes

Mode 1: Chronological List (Default)

Conversations in time order with groupings:

- Today
- Yesterday

- Last 7 Days
- Last 30 Days
- Older

Mode 2: Topic-Based Grouping

Conversations organized by health themes:

- Sleep & Recovery
- Digestion & Nutrition
- Stress & Mental Health
- Energy & Vitality
- Constitution & Balance
- Symptoms & Concerns
- Protocols & Practices
- General

2.4 Conversation Actions

Action	Trigger	Description
Open	Tap	View full conversation
Pin	Swipe right	Pin to top of list
Favorite	Swipe right	Mark as favorite
Share	Swipe left / Long press	Share with practitioner
Delete	Swipe left / Long press	Remove conversation
Add Notes	Long press	Add personal notes/tags
Export	Long press	Export as PDF/text/JSON
Continue	Long press	Resume thread in Today view

2.5 Search Capabilities

Basic Search:

- Matches keywords in titles, content, tags
- Filters by conversation type
- Filters by date range

Semantic Search (AI-Powered):

- Natural language queries
- "Show me conversations about digestion"
- "When did we discuss my stress?"
- "Find check-ins from last week"
- AI interprets intent and returns relevant results

2.6 Design Philosophy

Clean by Default, Powerful When Needed Minimal visual noise, rich functionality on demand. Simple list view hides powerful search, filtering, and management capabilities.

Smart Categorization Conversations intelligently grouped by type and theme. Users don't have to manually organize.

Flexible Discovery Multiple ways to find what you need — chronological browsing, topic exploration, or direct search.

Action-Rich Full set of conversation management tools accessible through intuitive gestures (swipe, long-press).

3. User Journey Breakdown

3.1 Viewing Chat History (Chronological)

Context: User opens Chat History from navigation.

Header Display:

```
┌──────────────────────────────────────────────────────┐
│  ←  Chat History                    [Grid/List]  ⚙️  │
└──────────────────────────────────────────────────────┘
```

Search Bar:

```
┌──────────────────────────────────────────────────────┐
│  🔍  Search or ask about conversations...         [×]│
└──────────────────────────────────────────────────────┘
```

Section Headers:

```
┌──────────────────────────────────────────────────────┐
│  TODAY                                            3  │
└──────────────────────────────────────────────────────┘
```

Conversation Cards:

```
┌──────────────────────────────────────────────────────┐
│  ✓ Morning check-in                               ›  │
│  7:30 AM   [Check-in]                                │
└──────────────────────────────────────────────────────┘

┌──────────────────────────────────────────────────────┐
│  🔍 Sleep pattern analysis                        ›  │
│  10:15 AM   [Analysis]                               │
└──────────────────────────────────────────────────────┘

┌──────────────────────────────────────────────────────┐
│  💬 Quick question about herbs                    ›  │
```

```
2:30 PM   [Chat]
```

Pinned Conversation (Special Styling):

```
📌 🔍 Constitutional deep dive                ›
Jan 15   [Analysis]
```

- Background: Warm cream (#FFF9F0)
- Border: Gold (#F8C471)
- Pin icon visible

3.2 Viewing Chat History (Topic-Based)

Context: User toggles to Grid/Topic view mode.

Topic Sections:

```
SLEEP & RECOVERY                                  8
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
✓ Morning check-in                                ›
🔍 Sleep analysis                                 ›
💬 Evening routine question                       ›
[View all 8 →]
```

```
DIGESTION & NUTRITION                             5
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
💬 Meal timing question                           ›
🔍 Agni assessment                                ›
[View all 5 →]
```

Topic Categories:

Topic	Description	Related Keywords
Sleep & Recovery	Sleep, rest, nidra, insomnia	sleep, rest, tired, insomnia, nidra
Digestion & Nutrition	Food, digestion, agni	digestion, food, eating, agni, ama
Stress & Mental Health	Anxiety, stress, emotions	stress, anxiety, mental, emotions
Energy & Vitality	Energy, fatigue, ojas	energy, tired, fatigue, ojas, prana
Constitution & Balance	Doshas, prakriti, vikriti	dosha, prakriti, vikriti, constitution

Symptoms & Concerns	Specific health issues	symptom, pain, issue, problem
Protocols & Practices	Routines, protocols, practices	protocol, routine, practice, dinacharya
General	Uncategorized conversations	(default)

3.3 Searching Conversations

Context: User taps search bar.

Active Search State:

```
┌──────────────────────────────────────────────────┐
│  🔍  digestion issues                          [×] │
└──────────────────────────────────────────────────┘
```

- Border: Teal (#4A9B8E)
- Focus ring visible

Filter Pills (Appear After Search):

```
┌──────────────────────────────────────────────────┐
│  [All]  Check-ins  Analyses  Chats  Clear filters │
└──────────────────────────────────────────────────┘
```

Results Header:

```
┌──────────────────────────────────────────────────┐
│  12 results for "digestion"              [Clear]  │
└──────────────────────────────────────────────────┘
```

Semantic Search Examples:

Query	Interpretation	Results
"Show me conversations about digestion"	Find all digestion-related	Conversations with digestion keywords/topics
"When did we discuss my stress?"	Find stress conversations with dates	Stress conversations with timestamp emphasis
"Find my morning check-ins"	Filter to morning check-in	All morning check-in conversations
"Conversations from last week"	Date filter	Past 7 days

3.4 Swipe Actions

Swipe Right (Pin & Favorite):

```
┌──────────────────────────────────────────────────┐
│  [Pin] [Favorite]     Morning check-in          › │
└──────────────────────────────────────────────────┘
```

Action	Color	Icon	Description
Pin	Gold (#F8C471)	📌	Pin to top of list

Favorite	Light teal (#7AB8AC)	⭐	Mark as favorite

Swipe Left (Share & Delete):

```
Morning check-in      [Share] [Delete]
```

Action	Color	Icon	Description
Share	Teal (#4A9B8E)	↗️	Share with practitioner
Delete	Coral (#E89B7F)	🗑️	Delete conversation

3.5 Long Press Menu

Context: User long-presses on a conversation card.

Menu Display:

```
Morning check-in
──────────────────────────────────────────
↗️  Open conversation
📌  Pin to top
⭐  Add to favorites
🏥  Share with practitioner
📝  Add notes/tags
💾  Export conversation
↩️  Continue thread
🗑️  Delete
──────────────────────────────────────────
Cancel
```

3.6 Share with Practitioner

Context: User selects "Share with practitioner" action.

Share Modal:

```
Share Conversation                    [×]
──────────────────────────────────────────
```

```
Select practitioner:

○ Dr. Sarah Chen
○ Dr. Raj Patel
○ + Add new practitioner

Add note (optional):

☑ Include my current metrics
☑ Include attached images

          [Cancel]    [Share]
```

Share Options:

- Select practitioner (from connected list)
- Add optional note
- Include current health metrics
- Include attached images

3.7 Add Notes & Tags

Context: User selects "Add notes/tags" action.

Notes Modal:

```
Add Notes & Tags                                    [×]
═══════════════════════════════════════════════════

Notes:

This was particularly helpful for
understanding my sleep issues...

Tags:
[Sleep] [Important] [Share] + Add tag

Suggested tags:
• Breakthrough  • Symptom relief  • Question
```

```
• Follow-up needed

          [Cancel]    [Save]
```

Tag Options:

- User-defined custom tags
- Suggested tags based on content
- Quick-add from suggestions

3.8 Export Conversation

Context: User selects "Export" action.

Export Modal:

```
Export Conversation                                  [×]
═══════════════════════════════════════════════════════

Format:
○ PDF (formatted)
○ Text file (.txt)
○ JSON (with metadata)

Include:
☑ Conversation messages
☑ Timestamps
☑ Attached images
☑ Associated metrics
☑ My notes and tags

          [Cancel]    [Export]
```

Export Formats:

Format	Use Case	Contents
PDF	Professional sharing, printing	Formatted conversation with branding
Text (.txt)	Simple backup	Plain text messages
JSON	Data portability	Full metadata, structured data

3.9 Continue Thread

Context: User selects "Continue thread" action.

Behavior:

1. Opens Today view with chat input
2. Pre-fills context: "Continuing from [conversation title]"
3. Shows link back to original conversation at top
4. Twin acknowledges continuation in response

Twin Response Example:

"I see you'd like to continue our discussion about your sleep patterns from January 15th. I remember we were exploring the connection between your elevated Vata and difficulty falling asleep. Where would you like to pick up?"

3.10 Delete Conversation

Context: User selects "Delete" action.

Confirmation Alert:

```
Delete this conversation?

This action cannot be undone.
The conversation will be permanently
removed from your history.

       [Cancel]    [Delete]
```

3.11 Empty States

No Conversations Yet:

```
                [Lotus Icon]

           No conversations yet

    Your journey with your Twin starts here!

         [Start First Conversation]
```

No Search Results:

```
[Search Icon]

No conversations found for
"digestion issues"

Try different keywords or
clear your filters

[Clear Search]
```

4. Technical Architecture

4.1 High-Level Architecture

```
PRESENTATION LAYER

ChatHistoryScreen

Header | Search | List | Modals
+ Toggle | + Filter | Views | (Actions)

STATE LAYER
```

```
│   ┌──────────────────┐   ┌──────────────────┐
┌──────────────────┐
│   │ Conversations    │   │  Search Store    │   │  History UI
│   │ Store            │   │                  │   │  Store
│   │                  │   │ • Query          │   │
│   │ • Conversations  │   │ • Results        │   │ • View mode
│   │ • Pinned         │   │ • Filters        │   │ • Scroll pos
│   │ • Favorites      │   │ • Suggestions    │   │ • Selected
│   │ • Topics         │   │ • Loading        │   │ • Modals
│   └──────────────────┘   └──────────────────┘
└──────────────────┘   │
└──────────────────────────────────────────────────────────────────────────────────────────────┘
──────────┘
                                   │
                                   ▼
┌──────────────────────────────────────────────────────────────────────────────────────────────
──────────┐
│                                 SERVICE LAYER
│
│   ┌──────────────────────┐
┌──────────────────────────────────────┐    │
│   │   Search Service     │   │     Conversation Service
│   │                      │   │
│   │ • Keyword search     │   │  • CRUD operations
│   │ • Semantic search    │   │  • Categorization
│   │ • Filter logic       │   │  • Topic assignment
│   │ • AI interpretation  │   │  • Pin/favorite management
│   └──────────────────────┘
└──────────────────────────────────────┘    │
│
│
```

```
│   ┌──────────────────────┐                                  │
┌──────────────────────────────────────────┐     │
│   │  Export Service      │  │      Share Service
│   │                      │  │
│   │                      │  │
│   │  • PDF generation    │  │  • Practitioner sharing
│   │  • Text export       │  │  • Metrics inclusion
│   │  • JSON export       │  │  • Access tracking
│   └──────────────────────┘
└──────────────────────────────────────────┘     │
└──────────────────────────────────────────────────────────────────────────────────────────┘
                                                 │
                                                 ▼
┌──────────────────────────────────────────────────────────────────────────────────────────┐
│                                DATA LAYER
│
│   ┌──────────────────────┐                                  │
┌──────────────────────────────────────────┐     │
│   │   Local Cache        │  │      Backend APIs
│   │                      │  │
│   │                      │  │
│   │  • Recent 50 convos  │  │  • Conversation service
│   │  • Search results    │  │  • Search service
│   │  • Topic groupings   │  │  • Export service
│   │  • User preferences  │  │  • Share service
│   └──────────────────────┘
└──────────────────────────────────────────┘     │
└──────────────────────────────────────────────────────────────────────────────────────────┘
```

4.2 Data Flow: Loading Chat History

```
User opens Chat History
        │
        ▼
┌──────────────────────────┐
```

```
┌──────────────────────┐
│ Check local cache    │ ── Return cached if fresh
└──────────────────────┘
        │
        ▼
┌──────────────────────┐
│ Fetch conversations:
│ • Recent (limit 20)
│ • Pinned (all)
│ • With metadata
└──────────────────────┘
        │
        ▼
┌──────────────────────┐
│ Process conversations:
│ • Categorize by type
│ • Group by date
│ • Assign topics
└──────────────────────┘
        │
        ▼
┌──────────────────────┐
│ Render list view
│ • Pinned at top
│ • Grouped by date
└──────────────────────┘
        │
        ▼
┌──────────────────────┐
│ Set up infinite      │ ── Load more on scroll
│ scroll listener      │
└──────────────────────┘
```

4.3 Data Flow: Semantic Search

```
User enters search query
        │
        ▼
┌──────────────────────┐
│ Debounce input       │ ── 300ms delay
│ (300ms)              │
└──────────────────────┘
        │
        ▼
┌──────────────────────┐
│ Determine search type:
│ • Keyword match?
│ • Natural language?
```

```
│ • Date reference?          │
└────────────────────────────┘
        │
        ▼
┌────────────────────────────┐      ┌────────────────────────────┐
│ If semantic query:         │ ────> │ Claude interprets          │
│ Send to AI                 │      │ query intent               │
└────────────────────────────┘      └────────────────────────────┘
        │                                          │
        │<─────────────────────────────────────────┘
        ▼
┌────────────────────────────┐
│ Execute search:            │
│ • Match keywords           │
│ • Filter by type           │
│ • Filter by date           │
│ • Filter by topic          │
└────────────────────────────┘
        │
        ▼
┌────────────────────────────┐
│ Rank results:              │
│ • Relevance score          │
│ • Recency weight           │
│ • User interactions        │
└────────────────────────────┘
        │
        ▼
┌────────────────────────────┐
│ Display results            │
│ • Highlight matches        │
│ • Show count               │
└────────────────────────────┘
```

4.4 Data Flow: Conversation Actions

```
User performs action (pin/favorite/delete/etc.)
        │
        ▼
┌────────────────────────────┐
│ Optimistic update          │ ── Update UI immediately
└────────────────────────────┘
        │
        ▼
┌────────────────────────────┐
│ Send API request           │
```

```
│
├── Success ──────────────────┐
                              ▼
                    ┌──────────────────────┐
                    │ Update local cache   │
                    └──────────────────────┘
│
└── Failure ──────────────────┐
                              ▼
                    ┌──────────────────────┐
                    │ Revert UI change     │
                    │ Show error toast     │
                    └──────────────────────┘
```

4.5 Data Flow: Export Conversation

```
User initiates export
        │
        ▼
┌──────────────────────┐
│ Show export modal    │
│ • Select format      │
│ • Select includes    │
└──────────────────────┘
        │
        ▼
┌──────────────────────┐
│ User confirms export │
└──────────────────────┘
        │
        ▼
┌──────────────────────┐
│ Fetch full convo     │ ── All messages, images, metrics
│ with all data        │
└──────────────────────┘
        │
        ▼
┌──────────────────────┐
│ Generate export:     │
│ • PDF: Format layout │
│ • TXT: Plain text    │
│ • JSON: Structured   │
└──────────────────────┘
        │
        ▼
```

```
Trigger download
or share sheet
```

4.6 Conversation Categorization Algorithm

```
CATEGORIZATION LOGIC:

Input: Conversation messages

Step 1: Analyze conversation start
  - Check if initiated by system prompt (check-in trigger)
  - Check message patterns

Step 2: Determine Type
  IF conversation started with:
    - Morning/evening check-in prompt → "check-in"
    - Assessment questions → "check-in"
  ELSE IF conversation contains:
    - Deep analysis (constitutional, health metrics) → "deep-
dive"
    - Protocol generation → "deep-dive"
    - Long-form responses with data → "deep-dive"
  ELSE
    - Default → "casual-chat"

Step 3: Assign Topics
  FOR EACH message in conversation:
    - Extract keywords
    - Match against topic keyword lists
    - Accumulate topic scores

  SELECT topics with score > threshold

Step 4: Return
  {
    type: "check-in" | "deep-dive" | "casual-chat",
    topics: ["sleep", "stress", ...]
  }
```

4.7 Search Ranking Algorithm

```
SEARCH RANKING:

Input: Query, Conversations
```

```
FOR EACH conversation:
  score = 0

  # Keyword match (40%)
  keyword_score = count_keyword_matches(query,
conversation) / total_keywords
  score += keyword_score * 0.4

  # Title match boost (20%)
  IF query in conversation.title:
    score += 0.2

  # Recency (20%)
  days_ago = (now - conversation.date).days
  recency_score = max(0, 1 - (days_ago / 90))
  score += recency_score * 0.2

  # User signals (20%)
  IF conversation.isPinned:
    score += 0.1
  IF conversation.isFavorited:
    score += 0.05
  IF conversation.hasNotes:
    score += 0.05

SORT conversations by score DESC
RETURN top results
```

5. AI Integration & Prompting Strategy

5.1 The Role of Claude in Chat History

Claude serves two primary roles in this section:

Capability	Description
Semantic Search	Interpret natural language queries to find relevant conversations
Thread Continuation	Understand context when resuming conversations

5.2 System Prompt for Semantic Search

```
You are a search assistant helping users find conversations
in their
chat history with their Ayurvedic Digital Twin.
```

YOUR ROLE

Interpret natural language search queries and translate them into
structured search parameters. Users may search for conversations
using informal language, and you need to understand their intent.

SEARCH INTERPRETATION

When given a search query, extract:

1. **Keywords** - Core terms to search for
2. **Topics** - Health topics the user is interested in
   - Sleep & Recovery: sleep, rest, tired, insomnia, nidra
   - Digestion & Nutrition: digestion, food, eating, agni, ama
   - Stress & Mental Health: stress, anxiety, mental, emotions
   - Energy & Vitality: energy, tired, fatigue, ojas, prana
   - Constitution & Balance: dosha, prakriti, vikriti, constitution
   - Symptoms & Concerns: symptom, pain, issue, problem
   - Protocols & Practices: protocol, routine, practice, dinacharya
3. **Date range** - If user mentions time periods
4. **Conversation type** - check-in, analysis, chat
5. **Special filters** - pinned, favorited, shared

EXAMPLES

Query: "Show me conversations about my digestion"
→ {
    keywords: ["digestion"],
    topics: ["digestion-nutrition"],
    dateRange: null,
    type: null
  }

Query: "When did we discuss my stress?"
→ {
    keywords: ["stress"],
    topics: ["stress-mental-health"],

```
    dateRange: null,
    type: null,
    includeTimestamps: true
  }

Query: "Find check-ins from last week"
→ {
    keywords: [],
    topics: [],
    dateRange: { start: "7_days_ago", end: "today" },
    type: "check-in"
  }

Query: "Conversations about sleep in January"
→ {
    keywords: ["sleep"],
    topics: ["sleep-recovery"],
    dateRange: { start: "2026-01-01", end: "2026-01-31" },
    type: null
  }

## OUTPUT FORMAT

Return a JSON object with search parameters.
```

5.3 System Prompt for Thread Continuation

```
You are the Ayurvedic Digital Twin for AyuAware. The user is
continuing a previous conversation.

## CONTEXT

The user is resuming a conversation from a previous session.
You have access to the full conversation history.

## YOUR ROLE

1. **Acknowledge the continuation** - Reference the previous
   conversation naturally
2. **Recall key points** - Summarize what was discussed
3. **Invite continuation** - Ask where they'd like to pick up

## APPROACH

- Be warm and welcoming
- Show you remember the conversation
```

```
- Summarize key points briefly (2-3 sentences)
- Ask an open question to continue

## EXAMPLE

Previous conversation: Discussion about sleep patterns and
Vata elevation

Your response:
"I see you'd like to continue our discussion about your sleep
patterns from January 15th. We were exploring the connection
between your elevated Vata and difficulty falling asleep, and
I suggested some evening routine adjustments. Have you had a
chance to try any of those, or would you like to explore
something else?"
```

5.4 Context Package for Search

```
SEARCH CONTEXT:

## User Query
- Raw query: {user_input}
- Previous searches: {recent_searches}

## Available Filters
- Types: check-in, deep-dive, casual-chat
- Topics: {topic_list}
- Date ranges: today, yesterday, last_7_days, last_30_days,
custom

## User Conversation Stats
- Total conversations: {count}
- Topics distribution: {topic_counts}
- Date range: {earliest} to {latest}

## Task
Interpret the query and return structured search parameters.
```

5.5 Context Package for Thread Continuation

```
CONTINUATION CONTEXT:

## Original Conversation
- Title: {conversation_title}
- Date: {conversation_date}
- Type: {conversation_type}
- Topics: {topics}
```

```
## Conversation Summary
{ai_generated_summary}

## Key Points Discussed
{bullet_points_of_key_topics}

## User Context
- Current Prakriti: {prakriti}
- Current Vikriti: {vikriti}
- Active protocols: {protocols}

## Task
Generate a natural continuation response that acknowledges
the previous conversation and invites the user to continue.
```

6. Data Models & State Management

6.1 Core Data Types

Conversation:

```
Conversation:
  - id: string
  - title: string
  - type: ConversationType
  - topics: string[]
  - createdAt: Date
  - updatedAt: Date
  - messageCount: number
  - lastMessage: string (preview)
  - isPinned: boolean
  - isFavorited: boolean
  - isCompleted: boolean (for check-ins)
  - hasUnread: boolean
  - notes: string | null
  - tags: string[]
  - sharedWith: string[] (practitioner IDs)
  - metadata: ConversationMetadata

ConversationType:
  - 'check-in' | 'deep-dive' | 'casual-chat'

ConversationMetadata:
  - checkInType: 'morning' | 'evening' | null
```

```
  - analysisType: 'constitutional' | 'health-metrics' |
'protocol' | null
  - associatedProtocol: string | null
  - associatedMetrics: string[] | null
```

ConversationMessage:

```
ConversationMessage:
  - id: string
  - conversationId: string
  - role: 'user' | 'assistant'
  - content: string
  - timestamp: Date
  - attachments: Attachment[]
  - metrics: AssociatedMetric[] | null

Attachment:
  - id: string
  - type: 'image' | 'document'
  - url: string
  - thumbnail: string | null
  - filename: string

AssociatedMetric:
  - type: string (CHI, NSL, etc.)
  - value: number
  - timestamp: Date
```

SearchQuery:

```
SearchQuery:
  - rawQuery: string
  - keywords: string[]
  - topics: string[]
  - dateRange: DateRange | null
  - type: ConversationType | null
  - isPinned: boolean | null
  - isFavorited: boolean | null

DateRange:
  - start: Date
  - end: Date

SearchResult:
  - conversation: Conversation
  - relevanceScore: number
  - matchedKeywords: string[]
  - highlightedTitle: string
```

```
  - highlightedPreview: string
```

TopicGroup:

```
TopicGroup:
  - id: string
  - name: string
  - icon: string
  - color: string
  - keywords: string[]
  - conversationCount: number
  - conversations: Conversation[]
```

ShareRequest:

```
ShareRequest:
  - conversationId: string
  - practitionerId: string
  - note: string | null
  - includeMetrics: boolean
  - includeImages: boolean
  - sharedAt: Date
  - viewedAt: Date | null
```

ExportRequest:

```
ExportRequest:
  - conversationId: string
  - format: 'pdf' | 'txt' | 'json'
  - includeMessages: boolean
  - includeTimestamps: boolean
  - includeImages: boolean
  - includeMetrics: boolean
  - includeNotes: boolean
  - generatedAt: Date
  - downloadUrl: string
```

6.2 State Structure

```
ChatHistoryState:
  # View State
  - viewMode: 'chronological' | 'topics'
  - currentView: 'list' | 'search-results' | 'topic-detail'
  - scrollPosition: number
  - isRefreshing: boolean

  # Conversations
  - conversations: Conversation[]
  - pinnedConversations: string[] (IDs)
  - favoritedConversations: string[] (IDs)
```

```
  - isLoadingConversations: boolean
  - conversationsError: Error | null
  - hasMore: boolean
  - offset: number

  # Topic Groups
  - topicGroups: TopicGroup[]
  - isLoadingTopics: boolean
  - expandedTopic: string | null

  # Search
  - searchQuery: string
  - searchResults: SearchResult[]
  - isSearching: boolean
  - searchError: Error | null
  - activeFilters: string[]
  - searchSuggestions: string[]

  # Selected Conversation
  - selectedConversation: Conversation | null
  - isLoadingConversation: boolean

  # Modals
  - shareModalOpen: boolean
  - shareModalConversation: string | null
  - notesModalOpen: boolean
  - notesModalConversation: string | null
  - exportModalOpen: boolean
  - exportModalConversation: string | null
  - deleteConfirmOpen: boolean
  - deleteConfirmConversation: string | null

  # Long Press Menu
  - longPressMenuOpen: boolean
  - longPressMenuConversation: string | null
  - longPressMenuPosition: { x: number, y: number }
```

6.3 State Management Approach

Zustand Stores:

```
Stores:

1. conversationsStore
   - Conversation list
   - Pinned/favorited tracking
```

```
   - CRUD operations
   - Pagination

2. searchStore
   - Search query and results
   - Filter state
   - Semantic search handling
   - Search history

3. topicsStore
   - Topic groups
   - Topic assignments
   - Expansion state

4. chatHistoryUIStore
   - View mode
   - Scroll position
   - Modal states
   - Long press menu
   - Loading states
```

7. Component Requirements

7.1 Header Components

ChatHistoryHeader

Navigation and controls:

- Back button
- Title "Chat History"
- View mode toggle (List/Grid icons)
- Settings button

ViewModeToggle

Switch between views:

- List icon (chronological)
- Grid icon (topics)
- Active state styling
- Animation on toggle

7.2 Search Components

SmartSearchBar

Search input:

- Search icon

- Input field with rotating placeholder
- Clear button (when active)
- Focus state with teal border
- Semantic search indicator

FilterPills

Category filters:

- Horizontal scrolling
- All / Check-ins / Analyses / Chats
- Active state styling
- Clear filters option

SearchResultsHeader

Results count:

- Result count and query
- Clear search link

7.3 List Components

SectionHeader

Date groupings:

- Section title (TODAY, YESTERDAY, etc.)
- Conversation count
- Uppercase styling

ConversationCard

Main list item:

- Category icon (left)
- Title
- Timestamp
- Type badge
- Chevron (right)
- Left border color by category
- Status indicators (completed, unread)

PinnedConversationCard

Pinned item variant:

- Warm cream background
- Gold border
- Pin icon visible
- Same content structure

FavoritedIndicator

Favorite marker:

- Star icon
- Gold color
- Position in card

SwipeActionsLeft

Right-reveal actions:

- Pin button (gold)
- Favorite button (teal)
- Smooth animation

SwipeActionsRight

Left-reveal actions:

- Share button (teal)
- Delete button (coral)
- Smooth animation

7.4 Topic View Components

TopicSection

Topic group container:

- Topic header with count
- Conversation list (condensed)
- View all button
- Expandable

TopicConversationItem

Simplified card:

- Category icon (small)
- Title
- Chevron
- No timestamp/badge

7.5 Long Press Menu Components

LongPressMenu

Full action menu:

- Conversation title header
- Action items list
- Cancel button
- Backdrop blur

LongPressMenuItem

Menu item:

- Icon
- Label
- Delete item in red
- Tap state

7.6 Modal Components

ShareModal

Share with practitioner:

- Practitioner selector
- Note input
- Include options (checkboxes)
- Cancel/Share buttons

NotesTagsModal

Add notes and tags:

- Notes text area
- Current tags
- Add tag input
- Suggested tags
- Cancel/Save buttons

ExportModal

Export options:

- Format selector (PDF/TXT/JSON)
- Include options (checkboxes)
- Cancel/Export buttons

DeleteConfirmation

Delete alert:

- Warning message
- Cannot undo notice
- Cancel/Delete buttons

7.7 Empty State Components

EmptyConversations

No conversations:

- Lotus icon
- Title
- Description
- Start conversation CTA

EmptySearchResults

No search results:

- Search icon
- Query display
- Suggestion text
- Clear search button

7.8 Loading Components

ConversationSkeleton

Loading placeholder:

- Gray blocks
- Shimmer animation
- Card dimensions

LoadingMoreIndicator

Infinite scroll loading:

- Spinner
- "Loading more..." text

RefreshIndicator

Pull to refresh:

- Rotating lotus icon
- Scale animation

7.9 Scroll Components

ScrollToTopButton

Quick scroll:

- Floating button
- Up arrow icon
- Teal background
- Fade in/out on threshold
- Smooth scroll animation

7.10 Animation Requirements

Animation	Duration	Easing	Trigger
Card entrance	300ms	ease-out	List load
Card stagger	50ms	-	Between cards
Swipe reveal	200ms	spring	Swipe gesture
Swipe return	200ms	spring	Release
Long press menu	200ms	ease-out	Long press
View mode toggle	400ms	ease-out	Toggle tap
Search results	200ms	ease-out	Results load
Highlight match	300ms	ease-out	Search result
Modal appear	300ms	ease-out	Modal open
Modal dismiss	200ms	ease-in	Modal close

Pull to refresh	500ms	ease-out	Pull release
Scroll to top	500ms	ease-out	Button tap
Delete item	300ms	ease-out	Delete confirm
Pin item	300ms	ease-out	Pin action

8. API Requirements

8.1 REST Endpoints

Get Conversations:

```
GET /api/v1/conversations?
mode=chronological&limit=20&offset=0

Query Parameters:
- mode: 'chronological' | 'topics'
- limit: number (default 20)
- offset: number (default 0)
- filters[]: 'check-in' | 'deep-dive' | 'casual-chat'

Response:
{
  "conversations": [
    {
      "id": "conv_001",
      "title": "Morning check-in",
      "type": "check-in",
      "topics": ["sleep-recovery"],
      "createdAt": "2026-02-01T07:30:00Z",
      "updatedAt": "2026-02-01T07:45:00Z",
      "messageCount": 8,
      "lastMessage": "Great! Keep up with the evening
routine...",
      "isPinned": false,
      "isFavorited": false,
      "isCompleted": true,
      "hasUnread": false,
      "tags": [],
      "metadata": {
        "checkInType": "morning"
      }
    },
```

```
    ...
  ],
  "hasMore": true,
  "totalCount": 47,
  "pinnedCount": 2
}
```

Get Pinned Conversations:

```
GET /api/v1/conversations/pinned

Response:
{
  "conversations": [...]
}
```

Get Conversations by Topic:

```
GET /api/v1/conversations/topics

Response:
{
  "topics": [
    {
      "id": "sleep-recovery",
      "name": "Sleep & Recovery",
      "icon": "🌙",
      "color": "#4A9B8E",
      "conversationCount": 8,
      "conversations": [...] // First 3
    },
    ...
  ]
}
```

Get Single Conversation:

```
GET /api/v1/conversations/:id

Response:
{
  "conversation": {
    "id": "conv_001",
    "title": "Morning check-in",
    "type": "check-in",
    "topics": ["sleep-recovery"],
    ...
    "messages": [
      {
```

```
        "id": "msg_001",
        "role": "assistant",
        "content": "Good morning! How did you sleep?",
        "timestamp": "2026-02-01T07:30:00Z",
        "attachments": []
      },
      ...
    ]
  }
}
```

Search Conversations:

```
GET /api/v1/conversations/search?q={query}
&filters[]={filters}

Query Parameters:
- q: string (search query)
- filters[]: conversation types
- semantic: boolean (use AI interpretation)

Response:
{
  "query": "digestion issues",
  "results": [
    {
      "conversation": {...},
      "relevanceScore": 0.92,
      "matchedKeywords": ["digestion"],
      "highlightedTitle": "Discussion about <mark>digestion</
mark>",
      "highlightedPreview": "...your <mark>digestion</mark>
has been..."
    },
    ...
  ],
  "totalResults": 12,
  "searchInterpretation": {
    "keywords": ["digestion"],
    "topics": ["digestion-nutrition"],
    "dateRange": null,
    "type": null
  }
}
```

Pin Conversation:

```
POST /api/v1/conversations/:id/pin
```

```
Response:
{
  "success": true,
  "isPinned": true,
  "pinnedAt": "2026-02-01T12:00:00Z"
}
```

Unpin Conversation:

```
DELETE /api/v1/conversations/:id/pin

Response:
{
  "success": true,
  "isPinned": false
}
```

Favorite Conversation:

```
POST /api/v1/conversations/:id/favorite

Response:
{
  "success": true,
  "isFavorited": true
}
```

Unfavorite Conversation:

```
DELETE /api/v1/conversations/:id/favorite

Response:
{
  "success": true,
  "isFavorited": false
}
```

Update Notes/Tags:

```
PATCH /api/v1/conversations/:id/notes

Request:
{
  "notes": "This was helpful for understanding...",
  "tags": ["sleep", "important", "share"]
}

Response:
{
  "success": true,
```

```
  "notes": "...",
  "tags": [...]
}
```

Share Conversation:

```
POST /api/v1/conversations/:id/share

Request:
{
  "practitionerId": "prac_001",
  "note": "Please review this discussion...",
  "includeMetrics": true,
  "includeImages": true
}

Response:
{
  "success": true,
  "shareId": "share_001",
  "sharedAt": "2026-02-01T12:00:00Z",
  "practitionerName": "Dr. Sarah Chen"
}
```

Export Conversation:

```
POST /api/v1/conversations/:id/export

Request:
{
  "format": "pdf",
  "includeMessages": true,
  "includeTimestamps": true,
  "includeImages": true,
  "includeMetrics": true,
  "includeNotes": true
}

Response:
{
  "success": true,
  "exportId": "export_001",
  "downloadUrl": "https://...",
  "expiresAt": "2026-02-02T12:00:00Z",
  "format": "pdf",
  "fileSize": 245678
}
```

Continue Conversation:

```
POST /api/v1/conversations/:id/continue

Response:
{
  "success": true,
  "newConversationId": "conv_048",
  "continuationContext": {
    "originalTitle": "Sleep pattern analysis",
    "originalDate": "2026-01-15",
    "summary": "Discussed connection between Vata and
sleep..."
  }
}
```

Delete Conversation:

```
DELETE /api/v1/conversations/:id

Response:
{
  "success": true,
  "deletedAt": "2026-02-01T12:00:00Z"
}
```

8.2 WebSocket Events (Optional)

```
WebSocket Events:

conversation.new
{
  "conversationId": "conv_048",
  "title": "Evening check-in",
  "type": "check-in",
  "createdAt": "2026-02-01T20:00:00Z"
}

conversation.updated
{
  "conversationId": "conv_048",
  "updatedFields": ["messageCount", "lastMessage"],
  "messageCount": 5,
  "lastMessage": "..."
}

share.viewed
{
  "shareId": "share_001",
```

```
  "conversationId": "conv_001",
  "practitionerName": "Dr. Sarah Chen",
  "viewedAt": "2026-02-01T14:00:00Z"
}
```

9. Success Metrics

9.1 Engagement Metrics

Metric	Target	Measurement
Chat History visits/ week	> 3	Users opening section
Time in section	> 1 min/ visit	Average session duration
Conversations opened	> 2/visit	Conversations viewed
Return visits	> 50%	Users returning within 7 days

9.2 Search Metrics

Metric	Target	Measurement
Search usage	> 30%	Sessions with search
Search success rate	> 80%	Searches leading to conversation open
Semantic search usage	> 40%	Natural language queries
Zero results rate	< 10%	Searches with no results

9.3 Action Metrics

Metric	Target	Measurement
Pin rate	> 10%	Users who pin conversations
Favorite rate	> 15%	Users who favorite conversations
Share rate	> 5%	Conversations shared with practitioners
Export rate	> 3%	Conversations exported
Continue thread rate	> 20%	Threads continued

9.4 Organization Metrics

Metric	Target	Measurement
Topic view usage	> 25%	Users switching to topic view
Notes/tags usage	> 15%	Conversations with notes/tags
Filter usage	> 20%	Sessions with filters applied

9.5 Technical Metrics

Metric	Target	Measurement
List load time	< 500ms	Time to display list
Search response time	< 1s	Time to show results
Semantic search time	< 2s	AI-interpreted search
Infinite scroll load	< 500ms	Time to load more
Export generation	< 10s	Time to generate export
Error rate	< 1%	Failed operations

10. Implementation Considerations

10.1 Performance Optimization

Lazy Loading:

- Load 20 conversations at a time
- Infinite scroll for more
- Virtual scrolling for long lists

Caching Strategy:

Data	Cache Duration	Storage
Recent 50 conversations	Until refresh	Local DB
Search results	5 minutes	Memory
Topic groupings	1 hour	Memory
Full conversation	On-demand	Memory

Search Optimization:

- Debounce input (300ms)
- Cache recent searches

- Index conversation content locally

10.2 Offline Handling

When Offline:

- View cached conversations
- View cached search results
- Cannot perform actions (queue)
- Show offline indicator

When Back Online:

- Sync queued actions
- Refresh conversation list
- Update search index

10.3 Conversation Categorization

Auto-Categorization:

- Analyze conversation start patterns
- Check for structured prompts
- Analyze content depth
- Assign type automatically

Topic Assignment:

- Keyword extraction from messages
- Match against topic keyword lists
- Multiple topics possible
- Re-analyze on conversation update

10.4 Search Implementation

Keyword Search:

- Full-text search on title and content
- Stemming for better matches
- Highlight matched terms

Semantic Search:

- Send query to Claude for interpretation
- Extract structured parameters
- Execute filtered search
- Combine with keyword results

10.5 Export Generation

PDF Export:

- Formatted layout with branding
- Include metadata header
- Conversation messages with styling
- Images embedded
- Metrics tables

Text Export:

- Plain text format
- Clear message separation
- Timestamps inline
- No images (note attachments)

JSON Export:

- Full structured data
- All metadata
- Image URLs
- Associated metrics

10.6 Practitioner Sharing

Share Flow:

1. User selects conversation
2. Chooses practitioner
3. Configures share options
4. System creates share record
5. Practitioner notified
6. Track view status

Shared Content:

- Conversation messages
- Optional: current metrics
- Optional: attached images
- User's notes

10.7 Accessibility Requirements

VoiceOver/Screen Reader:

- Cards: "Check-in, Morning check-in, 7:30 AM, button"
- Swipe actions announced
- Long press menu items labeled
- Search input labeled

Keyboard Navigation:

- Tab through cards
- Enter to open
- Arrow keys in menus
- Cmd/Ctrl+F for search

Touch Targets:

- Minimum 44x44px
- Swipe sensitive area: 60px from edge
- Long press timeout: 500ms

Dynamic Type:

- Respect user text size
- Layout adjusts gracefully
- Max +3 size categories

Reduced Motion:

- Disable complex animations
- Instant state changes
- Keep essential feedback

10.8 Analytics Events

Event	Properties	Purpose
`chat_history_opened`	source	Track entry
`conversation_viewed`	conversation_id, type	Content interest
`search_performed`	query, results_count, is_semantic	Search usage
`filter_applied`	filter_type	Filter usage
`view_mode_changed`	from, to	View preference
`conversation_pinned`	conversation_id	Organization
`conversation_favorited`	conversation_id	Organization
`conversation_shared`	conversation_id, practitioner_id	Sharing
`conversation_exported`	conversation_id, format	Export usage
`conversation_continued`	conversation_id	Thread continuation
`conversation_deleted`	conversation_id	Deletion
`notes_added`	conversation_id, has_tags	Notes usage
`long_press_menu_opened`	conversation_id	Menu usage
`swipe_action_used`	action_type	Gesture usage

10.9 Testing Priorities

Unit Tests:

- Categorization logic
- Search ranking algorithm
- Topic assignment
- Filter logic
- Date grouping

Integration Tests:

- Conversation list loading
- Search flow (keyword and semantic)
- Pin/favorite operations
- Share flow
- Export generation

E2E Tests:

- Browse to conversation open
- Search to conversation open

- Pin conversation and verify position
- Share with practitioner
- Export and download

Manual QA:

- Swipe gesture smoothness
- Long press menu behavior
- Search result relevance
- Topic grouping accuracy
- Export file quality

Appendix A: Topic Keywords Reference

Topic Keyword Lists

Topic	Keywords
Sleep & Recovery	sleep, rest, tired, insomnia, nidra, wake, bed, dream, fatigue, energy, recovery, rejuvenation
Digestion & Nutrition	digestion, food, eating, agni, ama, appetite, meal, stomach, gut, nutrition, diet, hunger
Stress & Mental Health	stress, anxiety, mental, emotions, worry, calm, peace, mind, nervous, tension, relax
Energy & Vitality	energy, tired, fatigue, ojas, prana, vitality, strength, weak, exhausted, vigor
Constitution & Balance	dosha, prakriti, vikriti, constitution, vata, pitta, kapha, balance, imbalance
Symptoms & Concerns	symptom, pain, issue, problem, concern, ache, discomfort, condition, feel
Protocols & Routines	protocol, routine, practice, dinacharya, ritual, habit, schedule, morning, evening

Categorization Rules

Check-in Indicators:

- Contains "check-in" or "check in"
- Started with morning/evening prompt
- Contains structured assessment responses
- Short, routine-like conversation

Deep Dive Indicators:

- Contains "analysis" or "assessment"
- Discusses constitutional details
- Protocol generation present
- Long, detailed responses
- Multiple health metrics discussed

Casual Chat Indicators:

- Quick question format
- Short exchanges
- General advice seeking

- Informal tone
- Default category

Appendix B: Color Reference

Category Colors

Category	Primary Color	Background (10% opacity)
Check-ins	#4A9B8E (teal)	rgba(74, 155, 142, 0.1)
Deep Dives	#E89B7F (coral)	rgba(232, 155, 127, 0.1)
Casual Chat	#7AB8AC (light teal)	rgba(122, 184, 172, 0.1)

UI Colors

Element	Color
Background	#F5F7F6
Card Background	#FFFFFF
Header Background	#FFFFFF
Border	#E0EBE8
Text Primary	#1A2B26
Text Secondary	#5A6B66
Text Tertiary	#A8B5B1
Pinned Highlight	#FFF9F0
Favorite	#F8C471

Appendix C: Swipe Gesture Specifications

Swipe Right Actions

Distance	Action	Visual
0-40px	Preview	Buttons peek
40-80px	Reveal	Buttons fully visible
80px+	Auto-execute first action	Pin

Swipe Left Actions

Distance	Action	Visual
0-40px	Preview	Buttons peek
40-80px	Reveal	Buttons fully visible
80px+	Auto-execute first action	Share

Gesture Physics

Property	Value
Spring stiffness	300
Spring damping	20
Velocity threshold	500
Haptic feedback	On threshold cross

End of Document